

Project Report

Auto Ticket Classification using Flow Designer

Team ID:	SWTID-2026-5797
Project Name	Auto Ticket Classification using Flow Designer
Date	03 october 2026

1. INTRODUCTION

1.1 Project Overview

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields. When a student creates a ticket, only the Caller and Short Description are entered; a flow then assigns Category and Subcategory and sends a confirmation email to the caller. The solution is no-code and was built in phases: requirement analysis, backend configuration, flow automation, testing, and deployment.

1.2 Purpose

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

2. IDEATION PHASE

2.1 Problem Statement

IT staff currently review each request manually to identify the issue type, select the correct category and subcategory, and inform the caller about ticket creation. This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

2.2 Empathy Map Canvas

Summarized from the problem statement; the project document does not contain a full empathy map, so only documented points are listed.

Who	Pains (documented)	Gains / needs (documented)
Students and teachers (callers)	Need to be informed that their ticket was created; submit recurring issue types (Wi-Fi, projector, password, slow computer).	Instant acknowledgement by email after submitting a ticket; only Caller and Short Description to enter.
IT agents / helpdesk staff	Manual review of every ticket is time-consuming, error-prone and does not scale.	Automatic Category and Subcategory; reduced manual effort; improved routing.

2.3 Brainstorming

The project document records the approach chosen rather than a brainstorming session. Design decisions documented:

- Classify using keywords in the short description (no scripting or machine learning).
- Use ServiceNow Flow Designer so the solution is no-code, easy to maintain and scalable.
- Use a custom table (Incident WorkFlow) with dependent Category / Subcategory choices for standardized data.
- Send an automated email to the caller on ticket creation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Caller (student / teacher)	System behaviour
1. Report issue	Opens a new ticket; enters Caller and Short Description (e.g., "WiFi not working in library").	Incident WorkFlow record is created with an INC auto-number.
2. Classification	No action needed.	Flow triggers (Category is empty), matches keywords and updates Category and Subcategory.
3. Acknowledgement	Receives email "Your Request for the issue has been submitted."	Send Email action emails the caller.
4. Follow-up	—	Classified ticket is ready for IT staff routing (Assigned group / Assigned to, State).

3.2 Solution Requirement

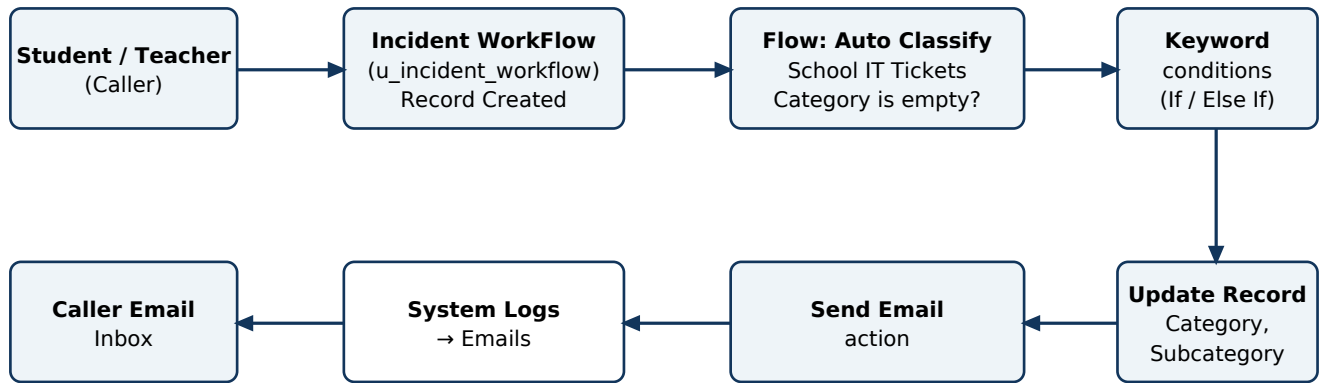
The system must:

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Send an automated email notification to the caller upon ticket creation.
- Store ticket information in a structured and standardized format.
- Ensure easy maintenance and future scalability.

Data fields of the Incident WorkFlow table:

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	—
Description	String	—
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

3.3 Data Flow Diagram



Short Description + Caller → classified record → confirmation email (logged in System Logs → Emails).

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Global application scope)
Automation	Flow Designer (Workflow Studio) - no-code flow with If / Else If logic and Update Record action
Data storage	Custom table Incident WorkFlow (u_incident_workflow) with auto-numbering (prefix INC, 5 digits)
Form configuration	Form Design; dictionary dependent field (Subcategory depends on Category)
Notification	Flow Designer Send Email action; verified via System Logs → Emails
Deployment	Local Update Set (Project Update Set) exported to XML

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Manual review and categorization of every ticket is time-consuming.	Flow classifies tickets automatically at creation.
Manual selection is error-prone.	Consistent keyword rules and dependent choice fields eliminate human error.
Callers must be informed about ticket creation.	Automated confirmation email to the caller.
Process is not scalable as requests grow.	No-code flow that is easy to maintain and extend.

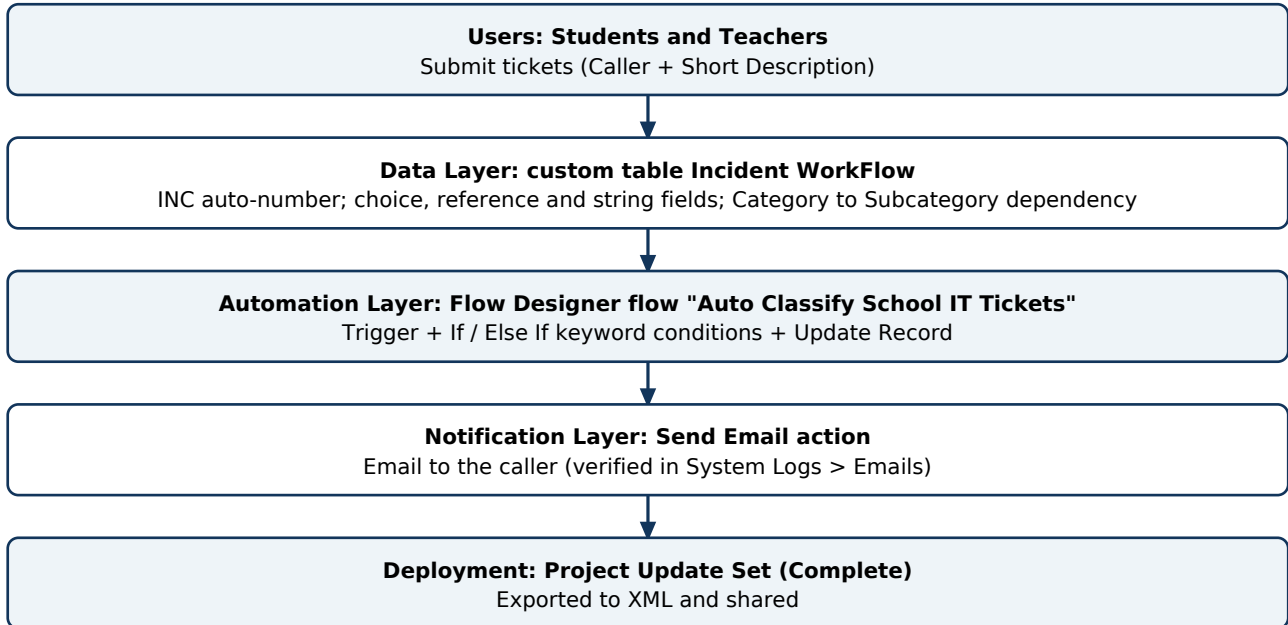
4.2 Proposed Solution

Trigger: Record Created on Incident Workflow where Category is empty. The flow evaluates the Short description and applies an Update Record action, then sends an email to the caller.

Keyword rule (overview)	Implemented condition (Short description contains)	Category	Subcategory
WiFi	Wi-Fi OR Network	Network	Wi-Fi
Projector	Projector	Hardware	Projector

Keyword rule (overview)	Implemented condition (Short description contains)	Category	Subcategory
Password or Login	Forgot password	Access	Forgot Password
Slow or Hanging	Slow computer	Performance	Slow Computer

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The solution was developed in phases (no dates or sprint estimates are recorded in the project document):

Phase	Activities
Phase 1 – Requirement Analysis & Planning	Business use case; creation of the Project Update Set (Global) and making it current.
Phase 2 – Backend Development & Configuration	Custom table Incident WorkFlow with auto-number; field creation and choices; Category-Subcategory dependency.
Phase 3 – Automation using Flow Designer & Email Notification	Flow “Auto Classify School IT Tickets”; trigger; Wi-Fi, Projector, Password and Slow computer classification; Send Email action; activation.
Phase 4 – Testing, Validation & Security	Data validation checks; Test Scenarios 1 and 2; email notification check. (Numbered Phase 6 in the source document.)
Phase 5 – Deployment & Final Documentation	Update set set to Complete; Export to XML; sharing the file.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

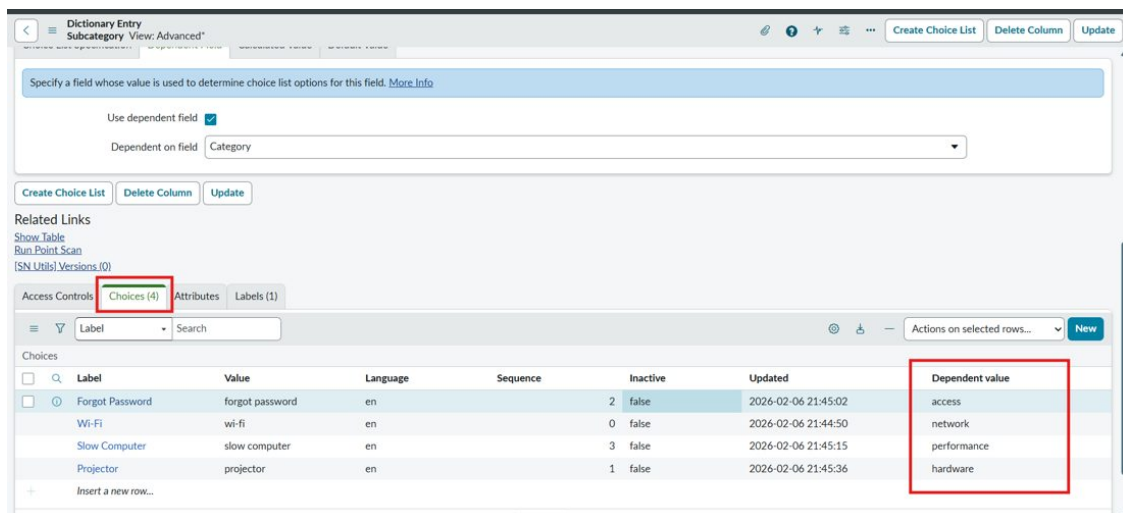
Data validation checks: mandatory fields captured correctly; auto-number generated without duplication; Category and Subcategory stored accurately; reference fields resolve to user and group tables.

Test Case ID	Scenario	Actual Result	Pass/Fail
FT-01	Text input (Caller, Short description)	Valid inputs accepted; tickets INC00511 and INC00512 created. Empty-field check not shown in document.	Pass (valid input)
FT-02	Auto-number	INC00511 and INC00512 generated uniquely.	Pass
FT-03	Auto classification	"WiFi not working in library" → Network / Wi-Fi; "Projector not turning on" → Hardware / Projector.	Pass
FT-04	Email notification	Email found in System Logs → Emails (send-ready, SMTP); preview displayed.	Pass
PT-01	Response time (target under 3 seconds)	Not measured in the project document.	Not tested
PT-02	Concurrent ticket creation	Not performed in the project document.	Not tested
PT-03	File upload load	No file-upload feature in this project.	N/A

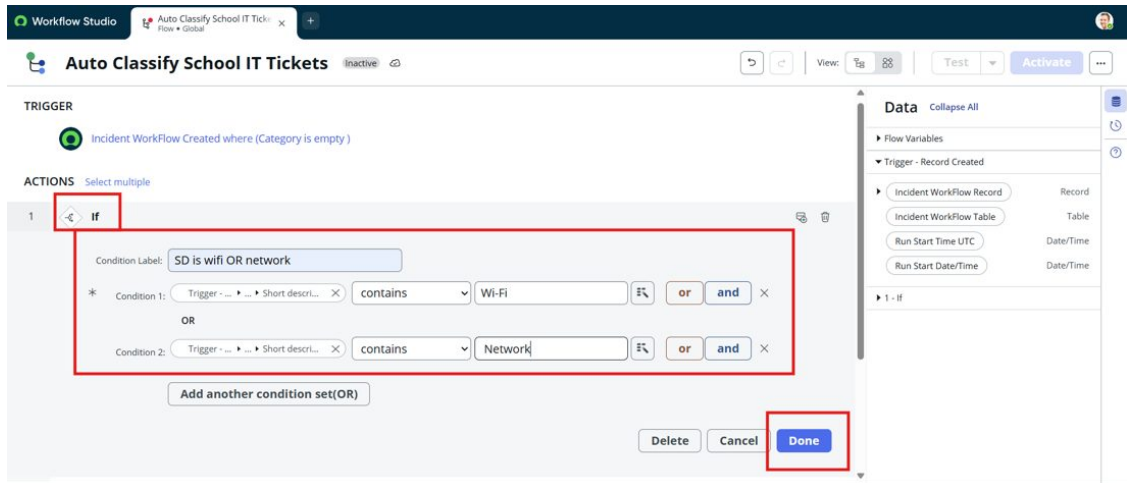
Accuracy: the solution is rule-based (no trained model), so training / validation accuracy and confidence scores are not applicable. Acceptance test summary: 9 test cases – 5 Pass, 0 Fail, 4 Not Tested; no defects recorded.

7. RESULTS

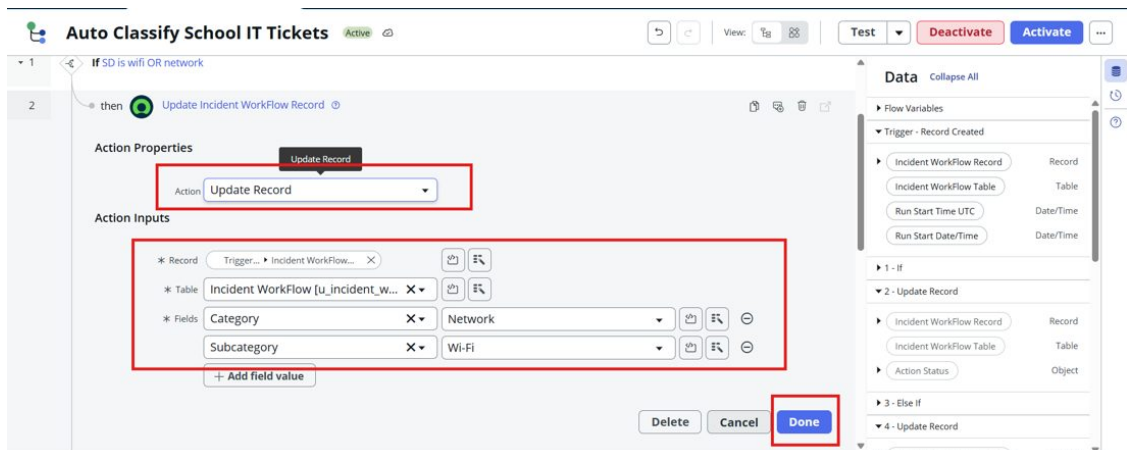
7.1 Output Screenshots



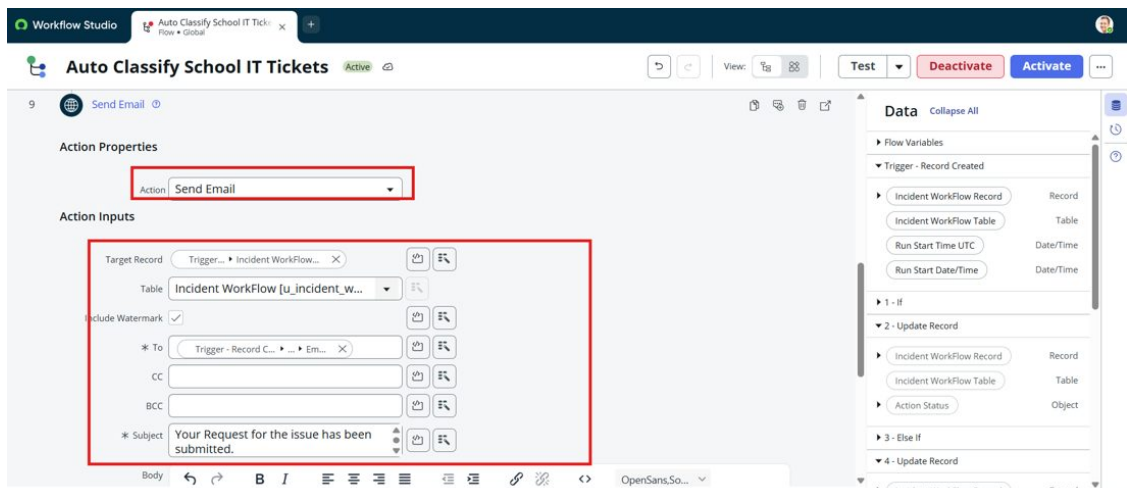
Subcategory choices with dependent values (Wi-Fi → network, Projector → hardware, Forgot Password → access, Slow Computer → performance).



Flow trigger: Record Created on Incident Workflow where Category is empty.



Update Record action for the Wi-Fi / Network condition (Category = Network, Subcategory = Wi-Fi).



Send Email action: To = Caller email; Subject "Your Request for the issue has been submitted."

Incident Workflow INC00511

Number: INC00511

* Caller: Alfonso Griglen

Category: Network

Subcategory: Wi-Fi

* Short description: My college Has issue with Network

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

Test Scenario 1: INC00511 auto-classified as Network / Wi-Fi.

Incident Workflow INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

* Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

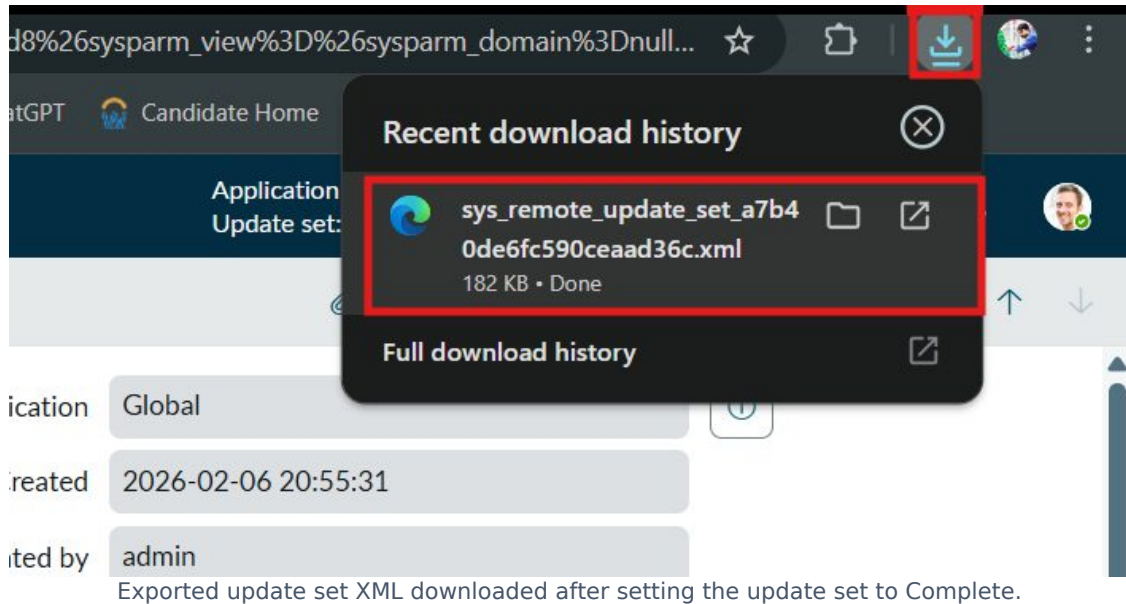
Test Scenario 2: INC00512 auto-classified as Hardware / Projector.

Created	Recipients	Subject	Type	Notification type	User ID
2026-02-06 23:04:59	savannah.kesich@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)
2026-02-06 23:04:52	eliseo.wice@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)

Email log (System Logs → Emails) showing the confirmation emails.



Preview of the email sent to the caller.



8. ADVANTAGES & DISADVANTAGES

Advantages

- Consistent classification that eliminates human error.
- Significantly reduces manual effort required by IT staff.
- Improves ticket routing efficiency and response efficiency.
- Instant email acknowledgement to the caller.
- No-code Flow Designer solution – easy to maintain and scalable, without complex scripting or machine learning.
- Structured, standardized data through choice fields and dependent choices.

Disadvantages / limitations (of the current implementation)

- Classification relies on keyword matching in the short description; it does not use machine learning or predictive intelligence.
- Covers four issue types (Wi-Fi, Projector, Password, Slow computer); other issues are not auto-classified.
- Described as well suited for small to medium environments such as educational institutions.
- Assignment automation and SLA tracking are not yet included.

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been achieved. Through structured requirement analysis, careful backend configuration and no-code automation, the system classifies IT tickets from the short description provided by students; dependent choice fields ensure data accuracy, and automated email notifications acknowledge ticket creation to the caller. The phased approach ensured clarity, stability and ease of maintenance, and testing confirmed reliable functioning and data integrity. The project demonstrates how ServiceNow can be used to build scalable, maintainable, user-friendly automation without complex scripting or machine learning.

10. FUTURE SCOPE

- Assignment automation.
- SLA tracking.

- Predictive intelligence.

11. APPENDIX

Item	Details
Source Code (if any)	No-code solution built in Flow Designer; no source code. Deliverable: Project Update Set exported as XML (sys_remote_update_set_a7b40de6fc590ceaad36c.xml, 182 KB).
Dataset Link	Not applicable – no dataset is used.
GitHub & Project Demo Link	Not provided in the project document.